THE ODIN PROJECT · FOUNDATIONS

JavaScript Basics



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01 Lessons in JavaScript Basics

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02 Variables, operators, data types, conditionals

let and const – const cannot be reassigned, let can; we do not use var

Operators – + - * / % ** and ++ / --, careful with "5" + 2 = "52"

8 data types – string, number, bigint, boolean, undefined, null, symbol, object

Comparison – always use === (strict) instead of ==

Conditionals – if / else if / else, switch, the ternary operator

Logical operators – && (and), || (or), ! (not)

basics.js

```
const name = "Aykhan";   // cannot change
let score = 0;           // can change
score++;                 // 1

"5" + 2;     // "52"  (string)
"5" * 2;     // 10    (number)
5 === "5";   // false (strict)

if (score >= 5) {
  console.log("You win!");
} else if (score === 0) {
  console.log("Start playing");
} else {
  console.log("Keep going");
}

const text = score > 0 ? "yes" : "no";
```

03

Tools, functions, problem solving, errors

Node.js and DevTools

Node runs JavaScript outside the browser. In Chrome DevTools I used the Console, the Elements panel and breakpoints in Sources to go through code line by line.

Function basics

Parameters and arguments, return values, function declarations, function expressions and arrow functions. Function scope keeps variables inside.

Problem solving

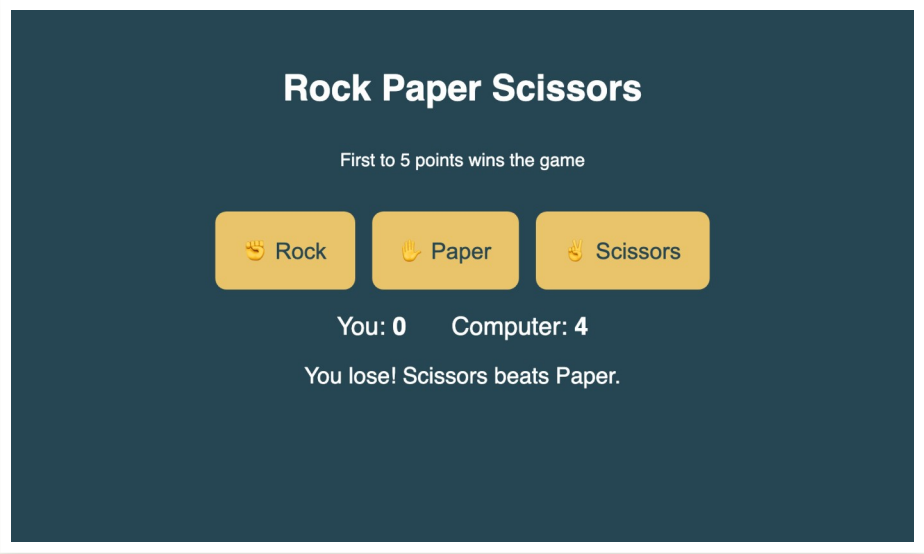
First understand the problem, then plan, write pseudocode and divide it into small steps. Then turn every step into code.

Understanding errors

Read the error message and the line number. ReferenceError: variable not defined. SyntaxError: wrong code. TypeError: wrong type of value.

04

Project: Rock Paper Scissors



- Computer choice with Math.random()
- playRound() compares the two choices
- Score for human and computer
- First to 5 points wins the game
- UI with buttons (Revisiting RPS lesson)

Live ayxan1110.github.io/odin-rock-paper-scissors/

Code github.com/Ayxan1110/odin-rock-paper-scissors

05

Clean code, loops and arrays

Clean code – good names (camelCase), consistent indentation, comments explain why

Arrays – ordered lists, index starts at 0; push, pop, slice, splice, includes

Loops – for, while, and for...of to repeat code

Array methods – map, filter, reduce, forEach – shorter than loops

Used in my projects – for loop in Etch-a-Sketch, forEach in RPS, includes() in the Calculator

arrays.js

```
const numbers = [1, 2, 3, 4, 5];

for (let i = 0; i < numbers.length; i++) {
  console.log(numbers[i]);
}

// new array, every number doubled
numbers.map((n) => n * 2);
// [2, 4, 6, 8, 10]

// only even numbers
numbers.filter((n) => n % 2 === 0);
// [2, 4]

// add everything together
numbers.reduce((sum, n) => sum + n, 0);
// 15
```

06

DOM manipulation and events

DOM – the page as a tree of nodes that JavaScript can change

Select – querySelector() and querySelectorAll()

Create and add – createElement(), appendChild(), append()

Change – textContent, classList, style, setAttribute()

Events – addEventListener("click", …), the event object, bubbling

dom.js

```
const container =
  document.querySelector("#container");

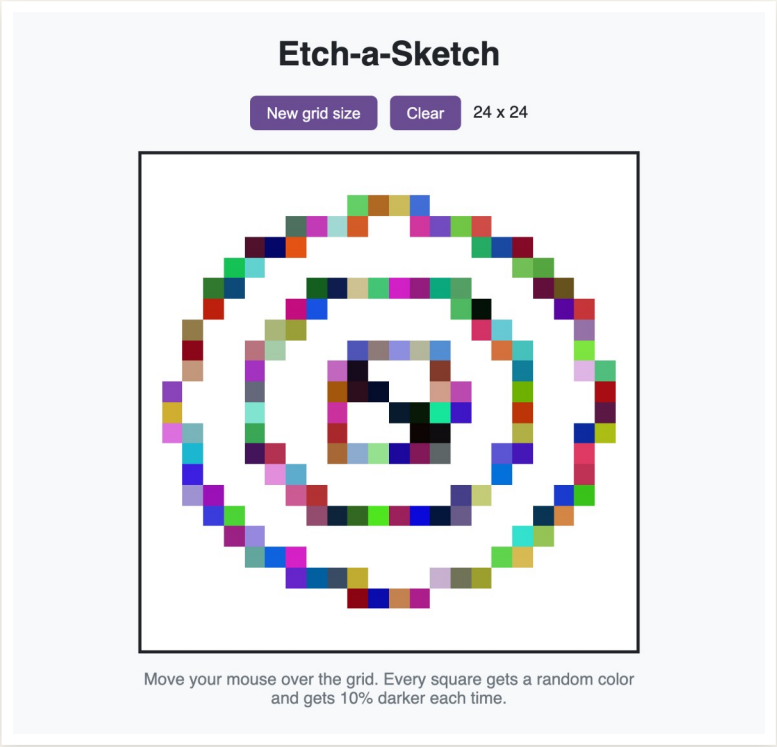
const button =
document.createElement("button");
button.textContent = "Click me";
button.classList.add("btn");
container.appendChild(button);

button.addEventListener("click", (event) => {
  console.log(event.target); // the button
  button.style.backgroundColor = "gold";
});

// many elements at once
document.querySelectorAll(".square")
  .forEach((sq) => sq.remove());
```

07

Project: Etch-a-Sketch



- 16 x 16 grid made with a JavaScript loop
- Flexbox with flex-wrap keeps it in a 480px box
- Squares change color on mouseenter
- Button with prompt() for a new size (max 100)
- Extra: random RGB and 10% darker every pass

Live ayxan1110.github.io/odin-etch-a-sketch/

Code github.com/Ayxan1110/odin-etch-a-sketch

08 Object basics

Objects – store data as key: value pairs

Add, change, delete – user.age = 23, delete user.age

Check a key – "name" in user, or loop with for...in

Reference – copying an object copies the reference, not the data

Array of objects – very common, e.g. a list of users or books

objects.js

```
const user = {
  name: "Aykhan",
  city: "Riga",
};

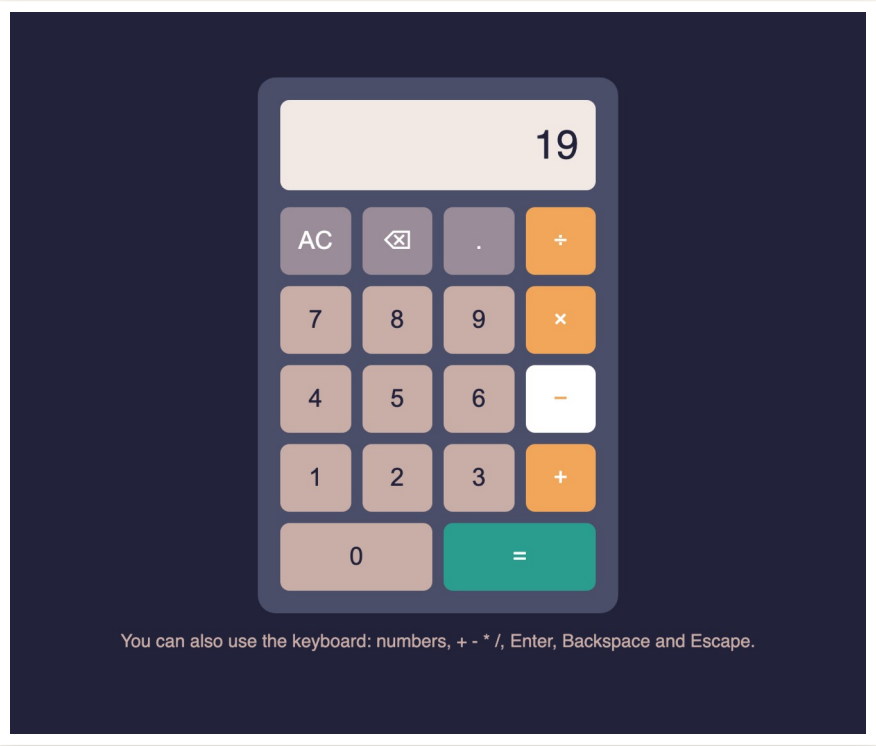
user.age = 23;           // add
user.city = "Baku";      // change
delete user.age;         // remove
"name" in user;          // true

const copy = user;       // same object!
copy.name = "Test";
user.name;               // "Test"

const users = [{ name: "A", age: 20 },
               { name: "B", age: 31 }];
users.find((u) => u.age > 30); // B
```

09

Project: Calculator



- add, subtract, multiply, divide + operate()
- One pair at a time: 12 + 7 − shows 19
- Rounds long decimals to fit the screen
- Error message when dividing by 0
- Extra: decimal, backspace, keyboard

Live ayxan1110.github.io/odin-calculator/

Code github.com/Ayxan1110/odin-calculator

10

Calculator: how the logic works

`firstNumber`

the number before the operator

`operator`

+ − × ÷ that was pressed

`currentInput`

the number on the display

Press	firstNumber	operator	currentInput	Display
1, 2	empty	empty	12	12
+	12	+	12	12
7	12	+	7	7
−	19	−	19	19
1, =	empty	empty	18	18

11 Projects on GitHub

Rock Paper Scissors	ayxan1110.github.io/odin-rock-paper-scissors/	github.com/Ayxan1110/odin-rock-paper-scissors
Etch-a-Sketch	ayxan1110.github.io/odin-etch-a-sketch/	github.com/Ayxan1110/odin-etch-a-sketch
Calculator	ayxan1110.github.io/odin-calculator/	github.com/Ayxan1110/odin-calculator

Thank you!